



Network Price List 2007/08

for

Standard Form Customer Connection Contract

effective

01 July 2007

Rev. 0

published

08 May 2007

Table of Contents

1. INTRODUCTION	4
2. RATES, FEES AND CHARGES	4
2.1 NETWORK PRICE LIST	4
2.2 NETWORK TARIFF STRATEGY STATEMENT	5
3. DEFINITIONS	5
3.1 MEANING OF WORDS	5
4. SECURITY	7
4.1 GENERAL	7
4.2 TYPES OF SECURITY	7
4.3 RESIDENTIAL CUSTOMERS	7
4.4 BUSINESS CUSTOMERS	8
4.5 REPAYMENT OF SECURITY PAID BY CASH, CHEQUE OR CREDIT CARD	8
4.6 DISCHARGE OR RELEASE OF SECURITY PAID BY GUARANTEE	8
4.7 DISCONNECTION OF SUPPLY	8
5. PRICING OPTIONS - DETERMINATION	9
5.1 DETERMINATION OF PRICING OPTIONS	9
5.2 DOMESTIC	9
5.3 CONTROLLED LOAD	10
5.4 DOMESTIC TIME OF USE	11
5.5 GENERAL SUPPLY NON TIME OF USE	11
5.6 GENERAL SUPPLY TIME OF USE	11
5.7 (SEASONAL) DEMAND TIME OF USE	11
5.8 TRANSITIONAL	12
6. PRICING OPTIONS - TYPES	13
6.1 TYPES OF PRICING OPTIONS	13
6.2 STANDARD NETWORK PRICING OPTIONS	13
6.3 COMBINATION PRICING OPTIONS	13
6.4 UNMETERED SUPPLY PRICING OPTIONS	13
6.5 TYPE 5 PRICING OPTIONS	13
6.6 SITE-SPECIFIC PRICING OPTIONS	13
6.7 EMBEDDED GENERATORS	14
7. NETWORK ACCESS CHARGES	14
8. DEMAND CHARGING	14
8.1 DEMAND METERING	14
8.2 DEMAND CHARGE CALCULATION	15
8.3 DEMAND PRO-RATION RULES	16
9. NETWORK ACCESS CHARGE AND CONSUMPTION PRO-RATION RULES	16
9.1 GENERAL CONSUMPTION PRO-RATION RULES	16
9.2 SPECIFIC CONSUMPTION PRO-RATION RULES IN RELATION TO IBT	17
10. DISTRIBUTION LOSS FACTOR	18
11. NEW SOUTH WALES ENERGY SAVINGS FUND	18
12. GST	19
13. OPERATING PERIODS, TIME-OF-DAY AND SEASON DEFINITIONS	19
13.1 TIME-OF-DAY DEFINITIONS FOR DOMESTIC TIME OF USE (DOMESTIC ToU) PRICING OPTION	19
13.2 TIME-OF-DAY DEFINITIONS FOR DEMAND ToU, GENERAL SUPPLY ToU AND ALL OTHER ToU	19
13.3 NETWORK BILLING AND DAYLIGHT SAVING TIME	20

13.4	SEASON DEFINITIONS FOR ALL DEMAND ToU NETWORK PRICING OPTIONS	21
14.	TREATMENT OF IMPORT/EXPORT POWER FLOWS WITHIN THE CONTEXT OF OUR NETWORK USE OF SYSTEM SERVICES CHARGES	22
15.	NATIONAL METERING IDENTIFIER (NMI).....	22
16.	VOLTAGES OF SUPPLY	22
17.	CHARGES FOR MISCELLANEOUS SERVICES	23
17.1	APPLICATION OF MISCELLANEOUS FEES AND CHARGES.....	23
1.	<i>DISCONNECTION/RECONNECTION VISIT</i>	<i>24</i>
2.	<i>SPECIAL METER READING FEE</i>	<i>25</i>
3.	<i>METER TEST FEE</i>	<i>25</i>
4.	<i>CONVEYANCING INQUIRY</i>	<i>26</i>
5.	<i>CONTROLLED LOAD CONVERSION FEE</i>	<i>26</i>
6.	<i>RECTIFICATION OF ILLEGAL CONNECTION FEE</i>	<i>26</i>
	APPENDIX 1: INTEGRAL ENERGY NETWORK PRICING OPTIONS	25
	APPENDIX 2: CONTROLLED LOAD APPLIANCES, TERMS AND CONDITIONS	30

SECTION 1: GENERAL INFORMATION

1. INTRODUCTION

In this document "we", "us", "our" and "ours" refers to Integral Energy; "you", "your" and "yours" refers to you, the customer.

Words in *italics* are explained in the Standard Form Customer Connection Contract. That contract is available for download from our website at www.integral.com.au.

Alternatively, you can obtain a copy by calling our Customer Service staff on 131 002.

2. RATES, FEES AND CHARGES

The documents mentioned below can be found on our website.

If you have any questions in relation to network prices please contact:

Michael Fogarty

Commercial Manager
Engineering Performance

Telephone: (02) 9853 4144
Fax: (02) 9853 6036
E-mail: michael.fogarty@integral.com.au

David Otto

Costing and Billing Manager
Business Performance

Telephone: (02) 9853 6289
Fax: (02) 9853 6036
E-mail: david.otto@integral.com.au

2.1 Network Price List

We have compiled this Network Price List (**Price List**) to provide you with:

- (a) a description of rates relating to our provision of *connection services* and *network use of system services*;
- (b) the basis for calculation of rates for *connection services* and *network use of system services*; and
- (c) information relating to miscellaneous fees and charges (Section 4),
when your *property* is connected to our *distribution system*.

This Price List forms part of the Connection Contract and complies with:

- the IPART Determination on NSW Electricity Distribution Pricing and associated Rules made under the National Electricity Code (June 2004); and
- the Annexure 13 "Pricing Principles" of the June 2004 NSW Electricity Distribution Pricing Determination No.2 by the IPART; and
- the *Electricity Supply Act*.

For the purposes of this Price List the following public holidays are deemed to be non-*business days*: New Year's Day, Australia Day, Good Friday, Easter Monday, Anzac Day, Queen's Birthday, Labour Day, Christmas Day, Boxing Day, and other gazetted public holidays in New South Wales.

All other non-gazetted holidays, such as bank holidays and other local holidays are deemed to be *business days*.

Integral Energy reserves the right to declare additional holidays, for the purpose of charging for *network use of system services*.

2.2 **Network Tariff Strategy Statement**

We have also prepared a separate document, which sets out how we determine the prices in the Price List (**Network Tariff Strategy Statement**). That statement includes information on:

- (a) the level and structure of prices;
- (b) the standard of service provided;
- (c) the methodology used to derive prices and their cost basis; and
- (d) medium term directions for prices and standards of service.

The Network Tariff Strategy Statement is available for download from our website at www.integral.com.au

3. **DEFINITIONS**

3.1 **Meaning of words**

In this Price List, unless inconsistent with the context or subject matter:

"Accumulation Meter" and **"Accumulation Metering"** have the meaning given in the New South Wales Electricity Supply Industry Metrology Procedure for Type 5, 6 and 7 Metering Installations.

"CEO" means the Chief Executive Officer of Integral Energy or a person authorised by the Chief Executive Officer or the Board to act on that person's behalf.

"Connection Contract" means the *standard form customer connection contract* entered into between a customer and us.

"Controlled Load" has the meaning given in clause 5.3.

"DST" is Eastern Daylight Saving Time and applies for the period gazetted by the NSW Government for use as local time.

"Daily Threshold" has the meaning given in clause 9.2.

"Demand Charge" has the meaning given in clause 8.

"Distribution Loss Factor" has the meaning given in clause 10.

"EST" is Eastern Standard Time and applies for the period gazetted by the NSW Government for use as local time.

"Export Energy" has the meaning given in clause 14.

"Import Energy" has the meaning given in clause 14.

"IPART" means the Independent Pricing and Regulatory Tribunal of New South Wales.

"Inclining Block Tariff" has the meaning given in clauses 5.2 and 5.5.

"Interval Meter" and **"Interval Metering"** have the meaning given in the New South Wales Electricity Supply Industry Metrology Procedure for Type 5, 6 and 7 Metering Installations.

"Interval Metering Data" has the meaning given the National Metering Identifier (NMI) Procedure ME-GN059 issued by NEMMCO.

"Metrology Procedures" means the New South Wales Electricity Supply Industry Metrology Procedure for type 5, 6 and 7 metering installations.

"MOR" means one of the Market Operations Rule(s) No.1 to 5 of 2001.

"Network Tariff Strategy Statement" has the meaning given in clause 2.2.

"NEM" means the National Electricity Market.

"NEMMCO" means the National Electricity Market Management Company.

"NMI" means the National Metering Identifier assigned to a *supply point*.

"Normal Business Hours" has the meaning given in clause 17.1.

"Price List" has the meaning given in clause 2.1.

"Pricing Category" has the meaning given in clause 5.1.

"Pricing Options" are set out in clause 5 of this Price List, only one of which will apply at any time to the *connection services* and *network use of system services* provided at each separately metered *supply point*.

"Season" has the meaning given in clause 5.8.

"Time of Day" has the meaning given in clauses 13.1 and 13.2.

"Voltage of Supply" has the meaning given in clause 16.

SECTION 2: NETWORK PRICES

4. SECURITY

4.1 General

We:

- (a) may require you to pay *security* where the requirements set out in clauses 4.3 or 4.4 below are met; and
- (b) must repay, release or discharge the *security* to you in accordance with clause 4.5 or 4.6 below.

The amount of the *security* may be up to:

- (c) 1½ times the average quarterly account;
- (d) 1¾ times the average bi-monthly account; or
- (e) 2½ times the average monthly account.

You will not be paid interest on the *security*.

4.2 Types of security

We accept only the following types of *security*:

- (a) cash, cheque or credit card from residential or business customers; or
- (b) guarantees, including Department of Housing guarantees, from residential customers, and bankers' guarantees from business customers.

4.3 Residential customers

We may require *security* from a residential customer prior to connection only if that customer:

- (a) has left a previous supply address without settling an electricity retail bill owed to us or any other standard network distributor, the debt remains outstanding and the customer has refused and refuses to make any arrangement to pay that debt;
- (b) has been responsible for the illegal use of electricity within the previous two years; or
- (c) does not have a satisfactory credit history in our reasonable opinion or cannot demonstrate satisfactory credit history with another network distributor to our reasonable satisfaction, and we have offered the customer an instalment plan or other payment option (for example pay as you go instalments, direct debit) and the customer has refused, or failed to agree to the offer.

4.4 **Business customers**

We may require *security* from a business customer prior to connection only if the customer:

- (a) is a new business;
- (b) has been responsible for the illegal use of electricity within the previous two years; or
- (c) does not have a satisfactory credit history in our reasonable opinion or cannot demonstrate a satisfactory credit history with another network distributor to our reasonable satisfaction.

4.5 **Repayment of security paid by cash, cheque or credit card**

Security paid by cash, cheque or credit card, will be refunded when:

- (a) a residential customer has made on-time payment of bills for one year from the date of the first bill; or
- (b) a business customer has made on-time payment of bills for two years from the date of the first bill and the maintenance of a satisfactory credit rating in our reasonable opinion.

When this occurs, we will inform you, in writing, of the amount that is refundable, and will credit that amount to your account within 10 *business days*.

4.6 **Discharge or release of security paid by guarantee**

Security paid in the form of a guarantee, will be discharged or released when:

- (a) a residential customer has made on-time payment of bills for one year from the date of the first bill; or
- (b) a business customer has made on-time payment of bills for two years from the date of the first bill and has maintained a satisfactory credit rating in our reasonable opinion.

If you meet the above requirements, we will inform you that a guarantee is no longer required. We will inform you in writing, and within 10 *business days*, of the above requirements being met.

4.7 **Disconnection of supply**

If you request us to disconnect *electricity supply*, before your *security* has been refunded we will:

- (a) inform you in writing of the amount of the *security* held; and
- (b) pay the amount either to you or your account.

This must occur within 10 *business days* of you ceasing to have connection.

5. PRICING OPTIONS - DETERMINATION

5.1 Determination of Pricing Options

We assign you a **Pricing Option** from those described in clause 0 and set out in Appendix 1 and 2, when we commence supply to you under a Connection Contract with us. The assigned Pricing Option will depend on the annual energy consumption measured at the *supply point*, the supply voltage at the *supply point*, the method of connection to our *distribution system* and the type of meter(s) installed at your *property*.

The available Pricing Options are

- Domestic;
- Controlled Load;
- Domestic Time of Use;
- General Supply Non Time of Use;
- General Supply Time of Use; and
- Demand Time of Use (Low Voltage, High Voltage and Sub-transmission).

If you wish us to change the Pricing Option applicable to a particular point, which is separately metered at the *property*, then you must apply to us in writing giving us 30 days notice. Any change we decide to make to the Pricing Option will take effect from the date on which your meter is next read after we process your application. We reserve the right to review your Pricing Option in the event of load change or of any other reason provided we do so acting reasonably (see also Appendix 1). If we change your Pricing Option we will notify your *retail supplier*. The decision of the CEO shall be final and conclusive in determining the applicable Pricing Option.

If we agree to your request to change a Pricing Option to or from an **Controlled Load** Pricing Option more than once within a 12 month period, we may charge you or your *retail supplier* the Controlled Load conversion fee (listed in item 5 of the table in clause 16.1) for each additional change.

The amount that you or your *retail supplier* must pay us, is calculated by multiplying the appropriate GST-inclusive "per kWh" price by the amount of electricity consumed (based on our measurement or estimate of your consumption) at each separately metered *supply point*.

This amount is then multiplied by the **Distribution Loss Factor** (see clause 10).

In addition, you must pay the relevant GST-inclusive "per day" **Network Access Charge** (see clause 7, if any) for each day from the *actual start date* until your Connection Contract with us terminates.

5.2 Domestic

A **Domestic** Pricing Option only applies to *connection services* and *network use-of-system services* supplied to the *supply point*, which is predominantly used for one or more of the following purposes:

- (a) private dwellings;
- (b) boarding and lodging houses, being any house in which three or more persons exclusive of the family of the proprietor thereof are lodged for hire or reward from week to week or for more than a week;

- (c) retirement villages;
- (d) residential sections of nursing homes and hospitals;
- (e) residential sections of educational institutions;
- (f) children's homes;
- (g) approved baby health centres, day nurseries and kindergartens;
- (h) churches, mosques, temples etc., being buildings or properties which are used principally for public worship or partly for public worship and partly for educational purpose; or
- (i) approved caravan sites,

and where that point has an accumulation (basic or disc - type 6) meter or an interval meter that is read as an accumulation meter.

The Domestic Pricing Option, an **Inclining Block Tariff (IBT)**, comprises the following components, as detailed in Appendix 1:

- a Network Access Charge (fixed component), and
- two variable components consisting of
 - 1) a **First Block Rate**, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) up to and including 1,750 kWh per quarter; and
 - 2) a **Second Block Rate**, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) above 1,750 kWh per quarter.

We reserve the right to assign a Domestic Pricing Option to any new *supply point* used for the purposes (a) to (i) and fitted with accumulation metering.

5.3 Controlled Load

A **Controlled Load** Pricing Option only applies where electricity load is separately metered and controlled at a *supply point* with an accumulation meter.

- (a) **Controlled Load 1** applies where specified appliances are controlled by our equipment so that supply may not be available between 7:00am and 10:00pm; and
- (b) **Controlled Load 2** applies where our equipment controls approved specified appliances so that electricity is available for restricted periods not exceeding 17 hours in any period of 24 hours.

Switching times will be managed to minimise network investment and meet customer needs for the load being controlled.

When a customer with Controlled Load chooses another Pricing Option, the Controlled Load meter may be removed but the Controlled Load relay must remain in place. The Controlled Load relay will remain our property, and must not be removed without the written approval of the Manager, Network Review and Compliance.

Customers with a Controlled Load relay are entitled to a controlled load network price only if all of the following conditions are met:

- (c) Controlled Load consumption is separately metered using the same type of meter as the uncontrolled portion of a customer's load;

- (d) Controlled Load consumption and uncontrolled load consumption is always synchronously read, i.e. on the same day; and
 - (e) the Controlled Load is operated by our load control systems or time switches.
- This Pricing Option is applicable only to approved appliances as detailed in Appendix 2.

5.4 Domestic Time of Use

A **Domestic ToU** Pricing Option applies to a property, which is predominantly used for one or more of the purposes set out in clause 5.2 at a *supply point* with a time of use meter from which interval meter data is obtained. See also clause 6.5.

We reserve the right to assign a Domestic ToU Pricing Option to any new or existing *supply point* fitted with interval metering.

5.5 General Supply Non Time of Use

A **General Supply Non-ToU** Pricing Option applies to low voltage electricity used for any purpose other than Domestic, at a *supply point* with an accumulation meter or an interval meter that is read as an accumulation meter. See also clause 6.5.

The General Supply Non-ToU Pricing Option, an **Inclining Block Tariff (IBT)**, comprises the following components, as detailed in Appendix 1:

- a Network Access Charge (fixed component), and
- two variable components consisting of
 - 1) a **First Block Rate**, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) up to and including 2,500 kWh per quarter; and
 - 2) a **Second Block Rate**, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) above 2,500 kWh per quarter.

We reserve the right to assign the General Supply Non-ToU Pricing Option to any new supply point fitted with accumulation metering, with exceptions being made for those indicated in clause 5.2.

5.6 General Supply Time of Use

A General Supply ToU Pricing Option applies to low voltage electricity used for any purpose other than Domestic at a *supply point* with a time of use meter from which interval meter data is obtained.

We reserve the right to assign the General Supply ToU Pricing Option to any new or existing *supply point* fitted with interval metering.

5.7 (Seasonal) Demand Time of Use

A **Demand ToU** Pricing Option applies to electricity used for any purpose that includes a Demand Charge at each *supply point*. Customers shall formally notify Integral Energy when there is an anticipated change in demand at their site, for example where plant or apparatus is being commissioned or undergoing testing, or where load is transferred or rearranged.

Effective 01 July 2006, a Demand ToU Pricing Option applies to electricity used for any purpose at a *supply point* with a time of use meter, from which interval meter data is obtained, and where

- total electricity consumption, per financial year, equals or exceeds 160,000 kWh, or
- Electricity is supplied at a voltage level defined as either High Voltage or Sub-transmission, according to clause 14.

We reserve the right to assign a Demand ToU Pricing Option to any new or existing *supply point* fitted with interval metering, where any of the criteria mentioned above applies.

We understand it is responsibility of the *retail supplier* to monitor consumption at a given *supply point*, and to apply to Integral Energy Network in timely manner, regarding any change in Pricing Option, according to clause 5.1, and detailed in Appendix 1, Table 4.

Effective 01 July 2007, a Demand ToU Pricing Option, as applied by Integral Energy, will vary according to the month of the year in which the electricity is supplied (**Season**), with distinct Seasons defined for the purpose of charging demand. See also clause 12.4

5.8 Transitional

A **Transitional** Pricing Option applies to electricity used for any purpose, where Integral Energy decides to either

- assign a different Standard Pricing Option to an existing *supply point* already on a Standard or Combination Pricing Option according to clause 6.2 or 6.3, or
- assign a Standard Pricing Option to an existing *supply point* currently on a Site-specific Pricing Option according to clause 6.6,

and where, as a result of such a decision, the projected increase in overall charges for that *supply point* is deemed to be unreasonable by Integral Energy.

We reserve the right to assign a Transitional Pricing Option to any new or existing *supply point* at the beginning of a new financial year, and to discontinue that same Transitional Pricing Option at the end of the same financial year, in which case every *supply point* affected reverts to the applicable Standard Pricing Option the next day.

A Transitional Pricing Option is NOT available for any application for change in Pricing Option, according to clause 5.1

6. PRICING OPTIONS - TYPES

6.1 Types of Pricing Options

Pricing Options are categories of prices, including all combinations of Pricing Options (**Combination Pricing Options**), which are available to customers depending on the type of metering equipment that is installed at a *supply point* on the customer's *property*. The Pricing Options are described in this clause 0 and are set out in Appendix 1 and 2.

Those Pricing Options apply irrespective of whether the customer is an Integral Energy regulated retail customer, an Integral Energy retail customer on a negotiated *supply contract* or a customer who has entered a negotiated *supply contract* with another licensed electricity retailer.

6.2 Standard Network Pricing Options

Standard Network Pricing Options (as set out in Table 1 of Appendix 1) are applicable to *supply points* located in the *Integral Energy distribution district*, unless one of the Pricing Options described in clauses 6.3, 6.4 or 6.5 apply.

6.3 Combination Pricing Options

Combination Pricing Options (as set out in Table 2 of Appendix 1) are combinations of Pricing Options, which are applicable to points at which a combination meter is installed. A combination meter is one which can meter both a controlled load and normal domestic (or general supply) consumption as two distinct energy flows. Combination Pricing Options are formulated based on the prices, which would ordinarily be applicable to each component of the Combination Pricing Option. For example, NC01 (Domestic/Controlled Load 1) Pricing Option consists of Domestic and Controlled Load 1 charges. Sites with Type 5 metering will be allocated different network Pricing Options (see clause 6.5).

6.4 Unmetered Supply Pricing Options

Unmetered Supply Pricing Options (as set out in Table 3 of Appendix 1) are applicable to *supply points* that are not metered. Energy consumption is calculated using the appropriate algorithm in the applicable Metrology Procedure.

6.5 Type 5 Pricing Options

Type 5 Pricing Options (as set out in Table 1 of Appendix 1) are applicable to *supply points* with a Type 5 (manually read interval) meter installed. For Domestic customers, the appropriate Pricing Option is N705 Domestic ToU. For all other Low Voltage customers with a Type 5 meter installed, the N845 General Supply ToU Pricing Option is the appropriate equivalent.

6.6 Site-specific Pricing Options

We may assign a Site-specific Pricing Option to any *supply point* in circumstances such as, but not limited to:

- we wish to recover investment associated with stranded or dedicated assets, which may otherwise not be recovered under a Standard Network Pricing Option,
- we agree to assign a Site-specific Pricing Option following an application from the customer, or a *retail supplier* acting on their behalf, to be assigned a Site-specific Network Pricing Option.

Applications requesting a change to a Site-specific Network Pricing Option must be submitted at least six months in advance, and the *supply point* in question must be at High Voltage or Sub-transmission level according to clause 15, and satisfy at least one of the following criteria:

- (1) Electricity consumption must have been equal to or greater than 100 GWh in total for the 36 months preceding the application; or
- (2) Electricity consumption must have been equal to or greater than 40 GWh per annum in each of the two financial years preceding the application; or
- (3) Monthly peak demand must have been equal to or greater than 10 MVA for 24 of the 36 months preceding the application.

If the application is accepted, a Site-specific Pricing Option will be applied to the *supply point* in question from the beginning of the financial year following the date of approval of the application.

We reserve the right to reassign a Standard Pricing Option to a *supply point* as set out in clause 6.2, and effective from the beginning of the next *billing period*, if we find that the *supply point* no longer satisfies any of the aforementioned criteria.

6.7 **Embedded Generators**

Any *supply point* that connects a generator to our *distribution system* must have an active NUOS account associated with it, as we will invoice a regular Network Access Charge as set out in clause 7, irrespective of whether or not an Import of Energy, as set out in clause 13, occurs at such a *supply point* during any *billing period*.

In cases where a High Voltage or Sub-transmission *supply point* exists primarily to connect a generator to our *distribution system*, and if energy consumed at that same *supply point* is less than five per cent of the energy generated during any *billing period*, then we may apply a General Supply Time Of Use Network Pricing Option, as set out in clause 5.6, to that *supply point*.

However, if the *supply point* in question exceeds the level given above, for more than two months during any period of twelve months, we reserve the right to assign a Standard High Voltage or Sub-transmission Demand Time Of Use Network Pricing Option to it, effective from the beginning of the next *billing period*.

7. **NETWORK ACCESS CHARGES**

A Network Access Charge (NAC) is applicable to all customers. The NAC is applied as a fixed daily charge for each *supply point* to the Integral Energy *distribution system*, i.e. per National Metering Identifier (NMI). More than one NAC may apply per NMI if there is more than one Pricing Option applicable to that NMI.

8. **DEMAND CHARGING**

8.1 **Demand Metering**

A Demand Charge is applicable to all customers on a Demand Pricing Option in respect of the demand for electricity their *supply points* place on the *distribution system*.

Demand is treated as a component of the Data Stream of Interval Metering Data, in accordance with Section 4.1 of NEMMCO NMI Procedure ME-GN059v004.

For the purpose of this Price List, the following definitions are considered equivalent:

NEMMCO		Quadrants covered	Integral Energy	
Energy Flow Definition	NMI Master Channel Suffix		Energy (or Power) Flow Definition	Corresponding Load Angle ϕ in degrees
Import kWh	B	2, 3	Effective, generated	180°
Export kWh	E	1, 4	Effective, consumed	0°
Import kVARh	K	3, 4	Reactive, generated	Leading (Capacitive)
Export kVARh	Q	1, 2	Reactive, consumed	Lagging (Inductive)

For a particular NMI, the Demand Charge for a month is calculated based on the maximum demand (kVA) for every half-hour metering interval during that month.

Let NEEEXXXXXX be a NMI with i feeders.
 Let $E_1, \dots, E_i$ be the kWh channels for each feeder.
 Let $K_1, \dots, K_i$ be the leading kVARh channels for each feeder.
 Let $Q_1, \dots, Q_i$ be the lagging kVARh channels for each feeder.

Then the kVA for each interval (usually half an hour) is calculated as

$$kVA = m \times \sqrt{(\sum E_n)^2 + (\sum (Q_n - K_n))^2}$$

where m is the number of metering intervals in an hour (usually $m = 2$ within the **NEM**)

For *supply points* fitted with an 'EMAIL Q3' meter, the measurement of reactive energy is predefined in the configuration of the meter as the addition of lagging and leading components. Common practice is to identify this measurement as Q , in these instances the kVA for each half-hour interval is calculated as.

$$kVA = m \times \sqrt{(\sum E_n)^2 + (\sum Q_n)^2}$$

The resultant kVA for sites metered with an 'EMAIL Q3' meter is in real terms equivalent to

$$kVA = m \times \sqrt{(\sum E_n)^2 + (\sum (Q_n + K_n))^2}$$

For *supply points* covered by the Connection Contract, the difference in the above computations is not significant, as either Q or K (usually K) is nil. See clause 13 and 14 for further details.

8.2 Demand Charge Calculation

If you are on a Demand ToU Pricing Option, you must pay a monthly charge based on the maximum recorded kVA in each half hour interval during the month, which according to clause 12.2, falls into the time period defined as 'Peak' (**Demand Charge**).

The Demand Charge is calculated as follows:

$$D_c = D_m \times d$$

where

D_c = Demand Charge for the month

D_M = the maximum demand (measured in kVA) recorded in respect of the *supply point* for the month, during the 'Peak' time periods.

d = Demand rate (\$/kVA/month) according to the season.

The demand rates used to calculate this charge are set out in Appendix 1.

8.3 Demand Pro-ration Rules

If you change your *retail supplier* part way through the month, we will calculate the Demand Charge (based on your maximum demand) for the entire month for the *supply point* and then pro-rate the charge to each *retail supplier* accordingly.

The pro-rated Demand Charge in respect of each *retail supplier* for each month is calculated as follows:

$$D_P = D_M \times d \times (t/T)$$

where

D_P = pro-rated Demand Charge in respect of the *retail supplier* for the month

D_M = the maximum demand (measured in kVA) recorded in respect of the *supply point* for the entire month, during the 'Peak' time periods.

T = number of days in the entire month

t = number of days with the relevant *retail supplier* to be charged

d = Demand rate (\$/kVA/month)

Say a customer **C** transfers from *retail supplier* 1 (**R1**) to *retail supplier* 2 (**R2**) on the 11th of January. For C, maximum demand during the 'Peak' time periods for the entire month was 150 kVA, and the relevant Pricing Option's demand rate \$5.00/kVA/month.

R1 would be charged according to the following formula:

$$150 \text{ kVA} \times 5.00 \text{ \$/kVA} \times (10/31) = \$241.94$$

R2 would be charged according to the following formula:

$$150 \text{ kVA} \times 5.00 \text{ \$/kVA} \times (21/31) = \$508.06$$

9. NETWORK ACCESS CHARGE AND CONSUMPTION PRO-RATION RULES

9.1 General Consumption Pro-ration Rules

Where Integral Energy is allowed by IPART to vary certain charges and rates as listed in Appendix 1, those variations may become effective part way through a *billing period*. The amount which each customer must pay for consumption under existing charges and rates, and for consumption under new charges and rates is calculated on a pro rata basis.

The pro-rated amount, in respect of the applicable **Network Access Charge (NAC)** and Energy Rate for each part of the *billing period* (respectively after the increase becomes effective) is calculated as follows:

$$E_P = E_M \times e \times (t/T) + n \times t$$

where

E_P = pro-rated sum of NAC and Energy Rate in respect to the *billing period*

E_M = the total consumption (measured in kWh) recorded in respect of the *supply point* for the *billing period*

T = number of days in the *billing period*

t = number of days with the relevant NAC and Energy Rate to be invoiced

e = Energy rate (¢/kWh)

n = NAC (\$/day)

Say a customer **C** used to have a quarterly *billing period*, and an increase in the Energy Rate becomes effective on the 31st of a 92 day *billing period*. C's energy consumption for the entire *billing period* was 920 kWh. For C, the relevant Pricing Option's NAC is 0.21 \$/day before and 0.22 \$/day after the increase, and the energy rate is 6.32 ¢/kWh before and 6.83 ¢/kWh after the increase.

For the first 30 days, C would be charged according to the following formula:

$$920 \text{ kWh} \times 6.32 \text{ ¢/kWh} \times (30/92) + 0.21 \text{ \$/day} \times 30 = \$25.26$$

For the last 62 days, C would be charged according to the following formula:

$$920 \text{ kWh} \times 6.83 \text{ ¢/kWh} \times (61/92) + 0.22 \text{ \$/day} \times 61 = \$55.08$$

9.2

Specific Consumption Pro-ration Rules in Relation to IBT

For billing purposes, the consumption threshold under the IBT structure mentioned in clause 5.2 and 5.5 is converted to a daily basis (**Daily Threshold**).

If average daily consumption in the *billing period* is below the Daily Threshold:

$$E_C = E_A \times T \times P_1$$

If average daily consumption in the *billing period* is above the Daily Threshold:

$$E_C = [L_1 \times T \times P_1 + (E_A - L_1) \times T \times P_2]$$

where

E_C = consumption charge in respect to the *billing period*

E_A = average daily consumption for the *billing period* (kWh)

L_1 = Daily Threshold for the First Block (kWh)

T = number of days in the *billing period*

P_1 = Energy rate for the First Block (¢/kWh)

P_2 = Energy rate for the Second Block (¢/kWh)

Say a General Supply non-ToU customer consumed 5,400 kWh during a *billing period* of 90 days, averaging 60 kWh per day. The energy rate is 5.00 ¢/kWh for the First and 5.50 ¢/kWh for the Second Block, and the IBT threshold is 15,000 kWh per year, i.e. the Daily Threshold is 41.0959 kWh per day.

The consumption charge is then calculated as follows:

$$41.0959 \text{ kWh} \times 90 \times 5.00 \text{ ¢/kWh} + (60.0 - 41.0959) \text{ kWh} \times 90 \times 5.50 \text{ ¢/kWh} = \$ 278.51$$

10. **DISTRIBUTION LOSS FACTOR**

The **Distribution Loss Factor** (DLF) is used by a *retail supplier* to increase the customer's metered energy amount to account for electrical losses in the *distribution system*. The DLF codes and values (as set out in Appendix 1, Table 5) are those current as at the date of publication of this Price List. However, they are subject to change.

The DLF codes and values shown in this Price List do not include the transmission loss factors which are used in the market settlement of energy to take account of electrical losses in the transmission network.

11. **NEW SOUTH WALES ENERGY SAVINGS FUND**

Effective 01 July 2005, all Integral Energy Network Pricing Options contain a contribution to the NSW Energy Savings Fund, except for those where it is specifically stated that they are exempt, as in Appendix 1.

The Network Pricing Options shown in the present Integral Energy 2007/08 Network Price List include recovery of the annual contribution by Integral Energy to the NSW Energy Savings Fund at the same level as in 2006/07.

SECTION 3: APPLICATION

12. GST

Both GST inclusive and GST exclusive Network Rates are shown in Appendix 1. At the time of this publication the applicable GST was 10%.

13. OPERATING PERIODS, TIME-OF-DAY AND SEASON DEFINITIONS

13.1 Time-Of-Day Definitions for Domestic Time Of Use (Domestic ToU) Pricing Option

The application of our rates to customers on the Domestic ToU Pricing Option for 'Peak', 'Shoulder' and 'Off-peak' periods is based on the time periods detailed below.

During Eastern Standard Time (EST), and during Daylight Saving Time (DST):

Business Days

Peak: 1300-2000
Shoulder: 0700-1300 & 2000-2200
Off-peak: All other times.

[illegible]

Non-business Days

Shoulder: 0700-2200
Off-peak: All other times.

[illegible]

13.2 Time-Of-Day Definitions for Demand ToU, General Supply ToU and all other ToU

The application of our rates to customers on any ToU Pricing Option other than the Domestic ToU for 'Peak', 'Shoulder' and 'Off-peak' periods is based on the time periods detailed below.

During Eastern Standard Time (EST), and during Daylight Saving Time (DST):

Business Days

Peak: 1300-2000
Shoulder: 0700-1300 & 2000-2200
Off-peak: All other times.

[illegible]

Non-business Days

Off-peak: All other times.

[illegible]

13.3 Network Billing and Daylight Saving Time

In order to maintain the same time limits during both Eastern Standard Time (EST) and Daylight Saving Time (DST), billing data is adjusted by shifting the data forward an hour to accommodate for the time shift during DST.

This means that at the start of DST (2am on Sunday) there will be an hour of null data when the time is shifted forwards an hour from EST to DST. Also, data for the period 23:00 to 24:00 in EST will be recorded the following day for the period 00:00 to 01:00 DST.

When DST ends, the time will move back an hour and there will be two sets of hourly data for the period from 2am until 3am, one set generated in DST and the second set generated after the time shift in EST. This data is aggregated for the purposes of billing the "per kWh" charge, but not for Demand charge calculations.

The table below represents how the data is shifted for DST. The value in each cell (1 to 24) is the period of the day in EST.

	00:00 - 01:00	01:00 - 02:00	02:00 - 03:00	03:00 - 04:00	(etc)	20:00 - 21:00	21:00 - 22:00	22:00 - 23:00	23:00 - 24:00
EST	1	2	3	4	(etc)	21	22	23	24
DST day 1	1	2	NULL	3	(etc)	20	21	22	23
DST	24	1	2	3	(etc)	20	21	22	23
EST day 1	24	1	2 3	4	(etc)	21	22	23	24

The first row represents a normal EST day.

The second row represents day one of DST. Note that the first two hours of the day are the same as EST. At 2am, when DST begins, the data is shifted forward one hour, resulting in a null value for the period between 2 and 3am. Following that, all data is shifted forward one hour as compared to EST.

The third row represents a normal DST day. The data from the last hour of the previous day in EST is used as the data for the first hour of the following day in DST.

The final row of the table represents the day when DST switches back to EST. The first three hours are as per normal DST days, then at 3am EST begins, which means there is a time shift back one hour. Thus data is recorded for 2 - 3am DST and also for 2 - 3am EST. This data is aggregated for the purpose of billing the energy (per kWh) component of the network charge, but not for the Demand charge component. After 3am, data is recorded and billed as per normal for EST.

Note that while there is less total consumption during the first day of the DST period, this is made up for when the switch back to EST occurs.

%

13.4 Season Definitions for all Demand ToU Network Pricing Options

Effective 01 July 2007, the following definition of three distinct Seasons will apply to all Demand ToU Pricing Options, for the purpose of charging demand each month, namely

Summer (S)

01 November, 00.00h to 31 March, 24.00h

Winter (W)

01 June, 00.00h to 31 August, 24.00h

Other Months (O)

01 September, 00.00h to 31 October, 24.00h

01 April, 00.00h to 31 May, 24.00h

Month	July	August	September	October	November	December	January	February	March	April	May	June
Summer					S	S	S	S	S			
Other			O	O						O	O	
Winter	W	W										W

For the financial year 2007/08, Integral Energy will maintain Summer and Winter demand rates identical.

14. **TREATMENT OF IMPORT/EXPORT POWER FLOWS WITHIN THE CONTEXT OF OUR NETWORK USE OF SYSTEM SERVICES CHARGES**

In the situation where an end-use customer generates into, as well as consumes energy from, Integral Energy's *distribution system*, *network use of system services* charges apply to the energy consumed by the customer (**Export Energy**). The energy generated back into Integral Energy's *distribution system* (**Import Energy**) is not recognised for *network use of system services* purposes, unless it is covered under a specific agreement.

Network use of system services charges are based on Export Energy only, i.e. Import Energy will not be subtracted from the Export Energy.

This policy also extends to Demand Charges, i.e. **Import Demand** will not be subtracted from **Export Demand** when calculating *network use of system services* charges.

Furthermore, metering shall be configured so that reactive energy is measured only when associated with energy consumed by the customer (Export Energy).

15. **NATIONAL METERING IDENTIFIER (NMI)**

We issue a National Metering Identifier (NMI) for each site in accordance with Section 1.7 of NEMMCO NMI Procedure ME-GN059. As a general rule, each NMI will attract one network charge, i.e. one charge for network access and a coincident demand charge if there are multiple meters at the NMI site. There may be some exceptions to this rule in special circumstances (for example, customers with backup supplies), however those will be dealt with on a case by case basis.

16. **VOLTAGES OF SUPPLY**

We reserve the right to determine the voltage of supply for a particular customer based on the size and nature of the load to be connected. Voltage levels referred to in the prices are:

- (a) Low Voltage (LV) - nominally 240 / 415 V;
- (b) High Voltage (HV) - nominally 12.7 kV SWER, 11 or 22 kV; and
- (c) Sub-Transmission (ST) - 33, 66 or 132 kV.

SECTION 4: MISCELLANEOUS FEES AND CHARGES

17. CHARGES FOR MISCELLANEOUS SERVICES

In accordance with the June 2004 NSW Electricity Distribution Pricing Determination No.2 by the IPART, Distribution Network Service Providers can levy various charges for services relating indirectly to the conveyance of electricity.

We will levy charges for these services on customers or their *retail suppliers*. If we levy the *retail supplier* then the *retail supplier* must decide whether to absorb the charge or pass it through to the end use customer.

17.1 Application Of Miscellaneous Fees And Charges

Unless otherwise specified, the fees and charges set out below apply between 7.30am and 4pm on *business days* (**Normal Business Hours**). Additional costs may be incurred for work undertaken at the specific request of customers outside these times.

The CEO has the authority to waive any of the fees set out below in individual cases, dependent upon the circumstances.

Miscellaneous Fees and Charges

Description of Fee or Charge (including Rate Code)	Excl. GST	Incl. GST	Rate Code
1. Disconnection/Reconnection Visit (a) Where it is necessary for us to call at a customer's <i>property</i> for the purpose of disconnecting supply for breach by the customer of a customer supply contract or a customer connection contract, where the disconnection does not occur on that occasion (for example, because of non-payment of an amount due under the contract or non-payment or provision of a <i>security</i>, but we do not proceed with disconnection because the customer tenders acceptable payment or <i>security</i>), the personal visit fee is:	\$35.00	\$38.50	CDH3
(b) Where it is necessary to call at a customer's <i>property</i> for the purpose of disconnecting supply for breach by the customer of a customer supply contract or a customer connection contract, or where a retail supplier has requested that the supply to the customer be disconnected, following the procedures laid out in our <i>Standard Form Customer Connection Contract</i> (for example, because of non-payment of an amount due under the contract or non-payment or provision of a <i>security</i>, and the customer does not tender acceptable payment or security and disconnection occurs), a fee is payable as follows: (i) Disconnection, reconnection at fuse(s): (ii) Disconnection, reconnection at meter box: (iii) Disconnection, reconnection at pole top or pillar box: (Disconnection will occur at the pole top or pillar box where access is denied for disconnection, or where there is evidence that the customer has reconnected supply in breach of the <i>electricity laws</i> or of a customer supply contract, or a customer connection contract.) (iv) Reconnection outside normal business hours. (For safety reasons, a reconnection outside normal business hours is only available at a meter box.)	\$70.00 \$70.00 \$117.00 \$75.00	\$77.00 \$77.00 \$128.70 \$82.50	CDF3 CDM3 DS18 AC02

Description of Fee or Charge (including Rate Code)	Excl. GST	Incl. GST	Rate Code
2. Special Meter Reading Fee Where both the following apply, a charge for a special meter reading may be levied: (a) where a "special meter read" as defined in the <i>Market Operations Rule (NSW Rules for Electricity Metering) No. 3 of 2001</i> takes place (but excluding any special meter reading of Metering Installations Types 1-4, which is an Excluded Distribution Service); and (b) where physical access to a meter on a customer's premises has been sought on an occasion, and when access was sought on that occasion physical access to the meter could not be obtained. We will not levy a special meter reading charge in the following circumstances: (a) if a customer requests a special meter reading to verify the accuracy of an original meter reading undertaken by us and that the special meter reading confirms that the original meter reading was inaccurate; or (b) for a meter reading undertaken by us for the purpose of issuing a final account when the customer is vacating the <i>property</i>.	\$35.00	\$38.50	AM01 / AM02 / AM03 / AM04
3. Meter Test Fee Fee for the testing of a meter at the customer's request: We will not be required to test the meter if the customer refuses to pay us this fee in advance. If the meter is found to be inaccurate, we will refund any amount paid in advance and the customer will not be liable to pay this fee.	\$58.00	\$63.80	MT01

Description of Fee or Charge (including Rate Code)	Excl. GST	Incl. GST	Rate Code
4. Conveyance Inquiry Applies to the supply of information regarding the availability of electricity supply, the presence of our equipment, power lines etc for property conveyance. Desk inquiry fee: Field visit fee: * GST is not applicable	 \$29.00 \$58.00	 \$29.00* \$58.00*	 none none
5. Controlled Load Conversion Fee (a) For the first change to or from an <i>Controlled Load Pricing Option</i> within a 12 month period the fee is: (b) For each additional change to or from an <i>Controlled Load Pricing Option</i> within the same 12 month period the fee is:	 Free \$47.00	 Free \$51.70	 none DM02
6. Rectification of Illegal Connection Fee Applies for work to rectify wiring, work or connections which are unauthorised or which are in breach of the <i>electricity laws</i> or the customer's contract with us, and to restore service to our requirements. Fee to rectify illegal connections:	 \$175.00	 \$192.50	 CI03

Please note:

Other fees and charges may also be applicable under special circumstances. For example, there are fees and charges relating to contestable work, *connection services* and asset relocations. Please contact us on 131 002, or visit our web site www.integral.com.au.

APPENDIX 1: INTEGRAL ENERGY NETWORK PRICING OPTIONS

Table 1 - Standard Network Pricing Options

Standard Network Pricing Options (GST Inclusive & Exclusive) effective 1 July 2007

Integral Energy Standard Network Prices	Code	Network Access Charge		Energy Rate										Demand Rate - Summer & Winter		Demand Rate - Other Months	
				Non-Time Of Use, Block 1 ¢ / kWh		Non-Time Of Use, Block 2 ¢ / kWh		Time Of Use Peak ¢ / kWh		Time Of Use Shoulder ¢ / kWh		Time Of Use Off-Peak ¢ / kWh					
		\$ / day												Time Of Use Peak-only \$ / kVA / month		Time Of Use Peak-only \$ / kVA / month	
		Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST
Domestic (IBT)	N70	0.210	0.2310	6.016	6.6176	7.097	7.8067										
Controlled Load 1 (Off-Peak 1)	N50	0.020	0.0220	0.470	0.5170												
Controlled Load 2 (Off-Peak 2)	N54	0.020	0.0220	1.879	2.0669												
Domestic ToU, Type 5 Meter	N705	0.550	0.6050					9.634	10.5974	6.914	7.6054	2.334	2.5674				
Domestic ToU, Type 6 Meter	N706	0.230	0.2530					10.234	11.2574	6.809	7.4899	1.956	2.1516				
General Supply Non-ToU (IBT)	N90	0.210	0.2310	5.034	5.5374	5.808	6.3888										
General Supply ToU	N84	0.340	0.3740					8.619	9.4809	6.334	6.9674	1.667	1.8337				
General Supply ToU, Type 5 M.	N845	0.550	0.6050					9.753	10.7283	7.838	8.6218	2.453	2.6983				
Transitional General Supply ToU*	N89	6.800	7.4800					8.607	9.4677	5.014	5.5154	0.408	0.4488				
LV Demand ToU	N19	6.800	7.4800					2.391	2.6301	1.640	1.8040	0.398	0.4378	7.432	8.1752	7.078	7.7858
HV Demand ToU	N29	11.250	12.3750					1.743	1.9173	1.299	1.4289	0.357	0.3927	5.308	5.8388	5.055	5.5605
ST Demand ToU	N39	17.680	19.4480					1.664	1.8304	1.266	1.3926	0.332	0.3652	4.048	4.4528	3.855	4.2405

* = This is a Transitional NP according to clause 5.8, applicable to select large customers only - it is NOT available on application.

All energy rates - with the exception of Controlled Load 1 & 2 (**N50** and **N54**) - include recovery of the Energy Savings Fund contribution.

Table 2 - Combination Pricing Options

Combination Pricing Options - Network Prices (GST Inclusive & Exclusive)

effective 1 July 2007

(Applicable to Customers who have a combination meter - Please refer to clause 6.3.)

Integral Energy Standard Network Prices	Code	Network Access Charge		Energy Rate					
				Uncontrolled Non-Time Of Use, Block 1		Uncontrolled Non-Time Of Use, Block 2		Controlled Non-Time Of Use	
		\$ / day		¢ / kWh		¢ / kWh		¢ / kWh	
		Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST
Domestic (IBT) + Controlled Load 1	NC01	0.230	0.2530	6.016	6.6176	7.097	7.8067	0.470	0.5170
Domestic (IBT) + Controlled Load 2	NC02	0.230	0.2530	6.016	6.6176	7.097	7.8067	1.879	2.0669
General Supply IBT + Controlled Load 1 *	NC03	0.230	0.2530	5.034	5.5374	5.808	6.3888	0.470	0.5170
General Supply IBT + Controlled Load 2 *	NC04	0.230	0.2530	5.034	5.5374	5.808	6.3888	1.879	2.0669

All energy rates - with the exception of Controlled Load 1 & 2 (**N50** and **N54**) - include recovery of the Energy Savings Fund contribution.

Table 3 - Unmetered Supply Pricing Options

Unmetered Supply Pricing Options
(GST Inclusive & Exclusive) effective 1 July 2006

(Street Lighting Pricing Options applicable to Public Lighting installations only.)

Integral Energy Standard Network Prices	Code	Network Access Charge \$ / day		Energy Rate ¢ / kWh	
		Excl. GST	Incl. GST	Excl. GST	Incl. GST
Street Lighting	ENSL	0.000	0.0000	5.155	5.6705
Nightwatch	ENNW	0.000	0.0000	3.370	3.7070
Other unmetered supplies		Price on Application			

All energy rates include recovery of the Energy Savings Fund contribution.

Table 4 – Change in Network Pricing Option

According to the current *Integral Energy Network Tariff Strategy Statement*, the following changes are likely to be approved on application

	Currently applicable Network Pricing Option	Application for changing Network Pricing Option to ...
Domestic Customers (240 or 415 V)		
All Domestic (Residential) Customers	N70, NC01 or NC02 N70, NC01 or NC02	N705 N706
General Supply (Business and Small Industrial) Low Voltage Customers (240 or 415 V)	Currently applicable Network Pricing Option	Application for changing Network Pricing Option to ...
Yearly Total Consumption less than 160,000 kWh ('small')	N90, NC03 or NC04 N90, NC03 or NC04 N84 N845	N84 N19 N19 N19
Yearly Total Consumption equal or greater than 160,000 kWh ('large')	N90, NC03 or NC04 N84 N845	N19 N19 N19
High Voltage Customers (11 or 22 kV)	Currently applicable Network Pricing Option	Application for changing Network Pricing Option to ...
All High Voltage Customers	N90 N84	N29 N29
Sub-transmission Customers (132, 66 or 33 kV)	Currently applicable Network Pricing Option	Application for changing Network Pricing Option to ...
All Sub-transmission Customers	N90 N84	N39 N39

We reserve the right to decline applications for all changes in Network Pricing Option not explicitly indicated here.

Specifically, Transitional Network Pricing Options are NOT available on application.

Table 5 – Distribution Loss Factors

All Distribution Loss Factors and Values are effective from 01 July 2007, 00.00 h EST, and are subject to change.

General Use	0708 DLF Code	Approved 0708 DLF Value
132 kV Network	HNVL	1.0059
Transmission Substation	HSTS	1.0045
Subtransmission Network	HSTL	1.0058
Zone Substation	HHVT	1.0058
High Voltage Network	HHVL	1.0098
Distribution Substation	HLVT	1.0231
Low Voltage Network	HLVL	1.0203
Embedded Generator	HGEN	1.0000

(continued from previous page)

Site-specific / NMI	0708 DLF Code	Approved 0708 DLF Value
NEEE000005	HHY1	1.0187
NEEE000006	HTY5	1.0267
NEEE000011	HHY6	1.0187
NEEE000014	HTY7	1.0087
NEEE000032	HTY2	1.0109
NEEE000046	HTV2	1.0047
NEEE000049	HHV1	1.0169
NEEE000066	HTY4	1.0234
NEEE000506	HHY4	1.0107
NEEE000707	HHY5	1.0292
NEEE000748	HTX2	1.0172
NEEE000749	HTX3	1.0459
NEEE000750	HTX4	1.0191
NEEE000757	HTV3	1.0015
NEEE000758	HIC1	1.0294
NEEE000759	HIC1	1.0294
NEEE000760	HTV4	1.0197
NEEE000762	HTV4	1.0197
NEEE000764	HTV4	1.0197
NEEE000766	HTV4	1.0197
NEEE000768	HTV4	1.0197
NEEE000770	HTY3	1.0122
NEEE000771	HTY3	1.0122
NEEE000881	HTY9	1.0000
NEEE000888	HTY8	1.0324
NEEE001596	HHY3	1.0104
NEEE001632	HTY6	1.0000
NEEE001656	HTV1	1.0209
NEEE001813	HHY2	1.0083
NEEE001814	HHY2	1.0083
NEEE001885	HTY1	1.0100
NEEE001892	HTX1	1.0227
NEEE004637	HHY7	now HHVT
NEEE004639	HHY7	now HHVT
NEEEW00001	HTF1	1.0063
NEEEW00002	HTF1	1.0063
NEEEW04150	HTF2	1.0073
NEEEW04151	HTF2	1.0073
NEEEW04152	HTF2	1.0073
NEEEW04153	HTF2	1.0073
NEEEW04154	HTF2	1.0073
NTTTW0RX60	TT03	1.0000
NTTTW0RX70	TT03	1.0000
NTTTWWRGBA	HIC2	disconnected

APPENDIX 2: CONTROLLED LOAD APPLIANCES, TERMS AND CONDITIONS

Controlled Load 1

The Controlled Load 1 Pricing Option applies where specified appliances approved by the CEO are controlled by our equipment (see clause 5.3 of the Price List), so that supply may not be available between 7 am and 10 PM EST. Supply will be made available for selected periods between 10 PM and 7 am EST, as approved by the CEO.

Storage Water Heaters

In relation to a heating unit in a storage water heater, the following additional conditions must all be met:

- (a) the rated hot water delivery of the storage water heater is not less than 100 litres, unless otherwise approved by the CEO;
- (b) our equipment controls the supply of electricity to the heating unit in the storage heaters;
- (c) the operation of any booster heating units is controlled in such a way that simultaneous operation with the main heating units is not possible; and
- (d) unless otherwise approved by the CEO, heating units must be arranged as multiples of 4.8 kW in accordance with the following table:

Rated Hot Water Delivery (in Litres)	Number and Rating of Heating Elements
Up to and including 400	1 x 4.8 kW
Above 400 and not exceeding 630	2 x 4.8 kW
Above 630	As necessary to provide the full amount of heat in approximately 8 hours but in any case not more than 20 watts/litre rated hot water delivery.

NOTE: The above requirements may be varied where a Controlled Load element is provided as a booster for a solar water heater. Controlled Load elements are available to Domestic and General Supply small retail customers.

Other Appliances

In relation to swimming pool pumps, pool heating equipment, dishwashers, clothes dryers, washing machines and other appliances approved by the CEO (other than those described above) the following additional conditions must all be met:

- (a) an approved storage water heater or storage space heater is also installed and supplied at the same Controlled Load 1 Pricing Option;
- (b) each appliance is permanently connected to the fixed wiring;

- (c) all Controlled Load circuits originate at the meter board and are controlled by our equipment so that supply is available during specified Controlled Load hours; and
- (d) for pool heating, the equipment rating shall not exceed 520 watts per square metre of the water surface, unless approved by the CEO.

Noise Control

Local councils may impose conditions relating to the use or operation of equipment causing offensive noise. Air conditioners, swimming pool pumps and heat pump motors may be subject to such conditions and customers should consult the local council before arranging for such equipment to operate at night on Controlled Load.

Thermal Storage Space Heaters (Heat Banks)

In relation to thermal storage space heaters, the following additional conditions must be met:

- (a) the thermal storage space heaters are controlled by our equipment and the aggregate rating is not less than 3 kW, and
- (b) the property must utilise the Domestic Pricing Option.

Under Floor Heating

In relation to under floor heating, the following additional conditions must be met:

- (a) the under floor heaters are controlled by our equipment and the aggregate rating is not less than 3 kW, and
- (b) the property must utilise the Domestic Pricing Option.

Ice Storage Systems

Controlled Load 1 also applies to ice storage systems provided the unit is controlled by our equipment and the aggregate rating is not less than 3 kW.

Transfer Between Pricing Options

A switch that transfers equipment normally supplied as a Controlled Load to another Pricing Option is not permissible.

Existing Installations

Storage water heaters and thermal storage space heaters, previously approved for connection as a Controlled Load, will continue to be eligible for supply under the Controlled Load 1 Pricing Option.

Special conditions - single person and dual occupant aged person accommodation

Notwithstanding the rated hot water delivery requirements of the Controlled Load 1 Pricing Option, in the case of single and dual occupant aged person accommodation owned and controlled by the NSW Department of Housing, or some institution/charity as defined by the CEO, the minimum rated hot water delivery may be reduced in accordance with the following table:

Number of Occupants in Property	Minimum Hot Water Delivery Rating	Minimum Kilowatt Rating
1	80 litres	3.6 kW
2	125 litres	3.6 kW

Controlled Load 2

The Controlled Load 2 Pricing Option applies where specified appliances approved by the CEO are controlled by our equipment (see clause 5.3 of the Price List), so that electricity is available for restricted periods not exceeding 17 hours in any period of 24 hours.

The same terms, conditions and restrictions as listed for Controlled Load 1 are applicable for Controlled Load 2, with the following exceptions:

- a) The Controlled Load 2 Pricing Option can be applied to an electric heat pump with a minimum tank size of 250 litres, but that pump cannot be consequently transferred to the Controlled Load 1 Pricing Option.
- b) The Controlled Load 2 Pricing Option does not apply to Ice Storage Systems.
- c) Special conditions - single person and dual occupant aged person accommodation - the following table replaces the one applicable to Controlled Load 1:

Number of Occupants in Property	Minimum Hot Water Delivery Rating	Minimum Kilowatt Rating
1 or 2	80 litres	3.6 kW